Assignment 7 Solution

Q.1. What is the primary objective of projecting a fuzzy relationset R(A,B) onto its constituent sets A and B?

(a) To increase the dimensionality of the fuzzy set

(b) To decrease the dimensionality of the fuzzy set

(c) To add more elements to the fuzzy set

(d) None of the above

Answer: (b)

Q.2. When projecting a fuzzy relation set R(A,B) onto a fuzzy set A, what happens to the universe of discourse?

(a) It remains unchanged

(b) It expands

(c) It becomes irrelevant

(d) None of the above

Answer: (d)

Q.3. In the projection of a fuzzy relation set$R(A_1, A_2, A_3,....A_k)$ onto sets A1, A2, ..., Ak, what does 'k' represent?

(a) The number of elements in the Cartesian product of the universe of discourse

(b) The number of dimensions retained in the projection

(c) The total number of generic variables in the original fuzzy relation set

(d) None of the above

Answer: (b)

Q.4. What is the purpose of the cylindrical extension of a fuzzy set?

(a) To decrease the number of generic variables

(b) To decrease the dimensionality of the fuzzy set

(c) To increase the complexity of the fuzzy set

(d) Assign the original membership values across new dimension

Answer: (d)

Q.5. When taking the cylindrical extension of a fuzzy set, what happens to the number of generic variables?

(a) It remains the same

(b) It decreases

(c) It increases by two

(d) None of the above

Answer: (a)

Q.6. Which of the following properties holds true for fuzzy relations but not for crisp relations?

(a) Demorgan's law of union

(b) Associativity for union

(c) Absorption of complement for intersection

(d) None of the above

Answer: (c)

Q.7. Which property holds good for both crisp relations and fuzzy relations?

(a) Absorption of complement for union

(b) Law of contradiction

(c) Absorption of complement for intersection

(d) None of the above

Answer: (d)

Q.8. Which of the following best describes the extension principle?

(a) A method for converting fuzzy sets using a known mapping function

(b) A principle for defining fuzzy logic operations

(c) A method for calculating fuzzy set cardinality

(d) None of the above

Answer: (a)

Q.9. In the context of the extension principle, what does "$f^{1}(Y)$" represent?

(a) The inverse of the membership function

(b) The inverse of the mapping function

(c) The inverse of the universe of discourse

(d) None of the above

Answer: (b)

Q.10. What does the extension principle help achieve in fuzzy set theory?

(a) Transformation of membership values from one domain to another

(b) Reduction of fuzzy set complexity

(c) Elimination of conflicting membership values

(d) None of the above

Answer: (a)